Estimating the effects of visual search latency and RAN on word- and sentence-level reading comprehension

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Introduction. In this work, we investigate two predictors of reading ability, namely visual search (VS) latency and rapid automatized naming (RAN), and their effects on word- and sentence-level reading comprehension in German (WLRC and SLRC, respectively). These predictors are especially relevant for reading acquisition because they predict reading fluency [1-2], which is foundational in the development of reading comprehension [3]. To capture this development, we focus on the first three grades of elementary school and the lower levels of reading comprehension. Strong WLRC requires fast lexical access which is enabled by efficient decoding (the mapping of graphemes to phonemes) or sight word reading in more advanced readers [4]. SLRC subsumes the same components, in addition to supralexical processes. VS is a task where the goal is to find one or multiple targets in an array of visually presented stimuli. It requires visuo-spatial attention [5], which should facilitate decoding. RAN, the rapid sequential naming of highly familiar visual stimuli, also requires visuo-spatial attention but is also dependent on efficient lexical access [6]. **Methods.** The general methodology of this study is to use two non-reading task scores (VS, RAN) to predict two reading comprehension scores. We conducted the four tasks in elementary schools in the German-speaking part of Switzerland over the course of three years. For the analysis, we combined up to three sessions of participants (N=61) attending first to third grade. In the VS task, participants had to find the target symbol in a line of 19 symbols of the same category (letters, mirrored letters or geometric shapes). We conducted two RAN tasks (one with letters, the other digits) [7], and combined them into a composite RAN score. For the comprehension scores, we used two of the subtests of ELFE, a reading comprehension test for German [8]: a picture-matching task for WLRC and a cloze task (gap in the middle of the sentence) for SLRC. Both tasks were timed. **Analysis and results.** We first fitted two Bayesian linear mixed models (Model 1 and 2) with avg. VS latency (VS latency), RAN speed in terms of items per second, grade and their respective interactions as fixed effects and random effects for each participant per session, using a Poisson likelihood. We z-score standardized the continuous independent variables and used sliding contrasts for grade (grade 2 vs grade 1, and grade 3 vs grade 2, respectively). Model 1 had a word score in number of correctly marked words as dependent variable, Model 2 an analogous sentence score. We used regularizing priors for both models. We found main effects of naming speed, VS latency and the difference between 2nd and 1st grade on both WLRC and SLRC. Furthermore, we found an interaction between naming speed and the difference between grade 2 and grade 1 on SLRC, which we resolved with an additional model (Model 3). For this model, we substituted the RAN predictor, as well as its interactions with grade, with a predictor of RAN within each grade. The other model specifications remained the same as Model 2. Here, we found that the effect of naming speed on SLRC decreased with each grade. In addition we calculated Pearson correlation coefficients between the predictors and found a weak correlation between VS latency and RAN. **Discussion.** Our results confirm the finding that RAN speed is a predictor of reading comprehension [9], at least on the word and sentence level. The finding that the effect of naming speed decreases with grade highlights the relevance of RAN as a predictor of *early* reading comprehension. Our results furthermore extend previous research that indicates that VS performance predicts WLRC [10] to also include SLRC, meaning that the association between VS and reading ability may goe beyond the facilitation of decoding. While RAN and VS share some similarities, our findings suggest the difference between them is worth exploring to inform their use as predictors in clinical practice.

Table 1: Pearson correlation coefficients (ρ) for naming speed (RAN), visual search latency (VS) and grade.

	RAN	VS	Grade
RAN	1		
VS	-0.31	1	
Grade	0.65	-0.41	1

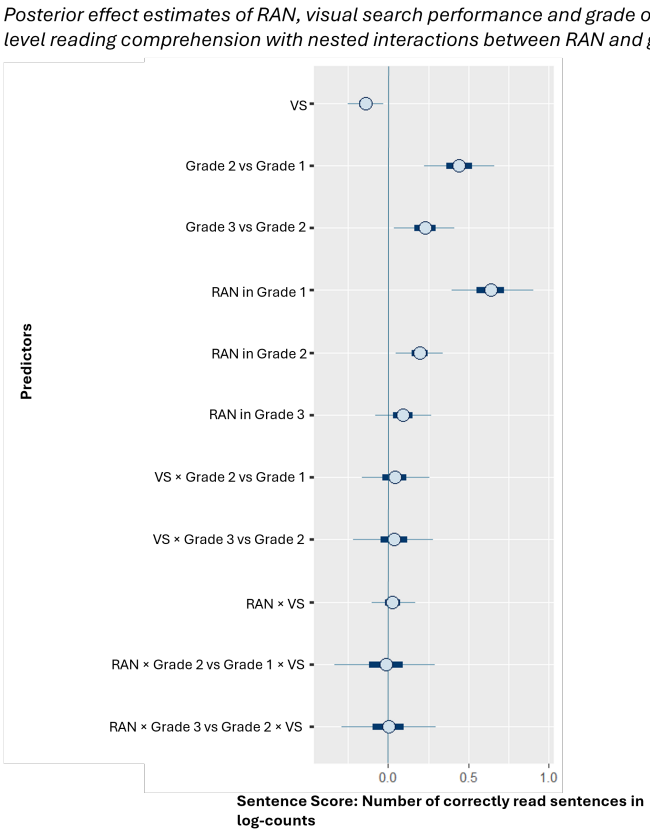

Figure 1: Effect estimates with 95% credible intervals for sentence-level reading comprehension in log-counts of correctly read sentences. The predictors are naming speed (RAN), grade (as a sliding contrast, to estimate the differences between adjacent grade levels), the effects of naming speed within the respective grades, and the interactions between naming speed, visual search latency and differences between grades.

References

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